



How We Tested

The entire test was divided into two procedures—one for testing printing capabilities and the other for the scanning capabilities of the MFD. The copier function was judged on two parameters—speed and quality. Faxing was considered a feature and not a performance criterion.

Our test bed comprised a Pentium 4 processor at 3.06 GHz, plugged on to an Abit 845GE board with 256 MB of Kingston 333 MHz DDR RAM, an MSI Ti4800 display card and a 40 GB 7200 rpm ATA IV Seagate Barracuda hard drive. We used Windows XP with SP1 as the operating system. The system was brought up-to-date, by loading all the latest chipset and graphics drivers. The USB patch for Windows XP was also installed, and the devices were connected on the USB 2.0 port for optimum performance.

Features: The main reason one would want to purchase an MFD is the numerous features it offers. So features formed the basis of our tests, and were awarded 15 per cent weightage. The various features that we logged for the printer aspect were the maximum printing resolution, paper capacity and type of interface offered. For the scanning aspect, features such as the maximum optical resolution supported, bit depth, scanner type and the sensor employed were noted. The MFD should work like a copier without a PC, so features such as one-touch copying, buttons for a colour or black and white copy, an option for multiple copies, and reduce and enlarge features were given prime consideration. After logging all of the above, weightages were allocated according to the importance of each particular feature.

Performance: Each of the two major test procedures—the printer and the scanner tests—reflected two prime parameters, speed and quality. For the printing speed test, we logged the time required for printing our standard black and white text document at normal settings, and for printing our test image file on photo paper, where the MFD was set to its best printing mode.



The Test page for Cost per page analysis, helped us make a buying decision

For determining the quality of the output text, we printed a document with varying point size text and looked for jarred edges and smudging. For evaluating image quality, we took the same printed photograph and looked at the parameters we decided upon as quality criteria. We used a standard combination document to judge the MFD's performance at printing documents containing both images and text.

This time around, we also included a cost-per-page test. For this, we prepared a test document with four paragraphs with the same content, but in four different colours—cyan, magenta, yellow and black. This ensured that each colour was used in the same proportion when the test page was printed. Before the test, we noted the ink level, and then printed out as many copies as could get printed before the ink level dropped to 20 per cent of the original.

For the scanning speed test, we logged the time taken for scanning a standard text document at a resolution of 200 dpi. The same test was repeated for a test photograph at resolutions of 150 dpi and 600 dpi. The time taken was noted.

To determine the scan quality, an assortment of tests were run on the MFDs. These included:

The image quality test using the IT8 card: An IT8 card is used by professionals all over the world to gauge the colour purity of a scanned image. This card was scanned at 300 dpi to check colour reproduction in the light and dark spectra. This test is of prime importance, since it reflects the scanner's accuracy in colour reproduction.

The image quality test using a resolution chart: In this test, a resolution chart consisting of several lines was scanned at 300 dpi; then, using the Magic Wand feature of Adobe Photoshop, all the scanned blocks were analysed. The more lines the MFD differentiated, in each of the five blocks, the more points it scored. This test essentially determined the MFD's ability to pick up fine details.

When not to buy an MFD

While MFDs are generally suitable for areas that lack in space, one might sometimes find MFDs lacking in performance. Here are some instances where buying individual peripherals would be a better option.

The Printer Unit: The quality of the printing units incorporated into MFDs have improved a lot over time, and today they give performance comparable to some of the best in the consumer to prosumer printers. But for business users and professionals in specific fields such as Web designing and publishing, printer units built in prosumer MFDs do not deliver the quality of a desktop printer. This segment requires photo-printer performance. Also, most MFDs come with support for A4 sizes—so for A3 size printing, you can't bank on a MFD.

The Scanner Unit: The scan quality offered by most MFDs is comparable to that of a standalone entry-level scanner. Most consumer MFDs offer quality not at par with that of entry-level scanners. Users who need the quality of a prosumer scanner may get a raw deal, since the cheapest MFD today costs about the same as a mid-range scanner, and one cannot, therefore, expect similar scan quality. Professional graphics designers, Web designers who require precise colour reproduction, and the like, just won't be satisfied with the results of a typical MFD's scanner. Such users would do well to avoid an MFD.

The Copier Unit: The cost for a copy taken on an inkjet MFD is higher than that of one taken on a regular copier. Also, an MFD takes longer for a copy. If your work demands include a large number of copies, it's wise to invest in a regular copier.